

Online Test 2 (10%)

- Due 19 Jun at 23:59
- Points 100
- Questions 24
- Available until 19 Jun at 23:59
- Time limit 90 Minutes

Instructions

Due Date: Week 7 Lab Session

Weighting: 10%

About this test

1. This is an online test worth 10% of the total marks for this unit.
2. It consists of 23 questions on theories and applications. Using the lab environment in the quiz is essential to find answers.
3. You may **only attempt this test once**.
4. **Do not navigate away from the test (close the browser tag) before you have completed it and submitted it.**
5. You have **90 minutes** to complete the test; exceeding this will result in a loss of marks and penalties.
6. This is a **close-book test**.
7. Once you begin the quiz, a summary will appear in the upper right corner, showing the number of questions (completed and total) and the time remaining.
8. Please bring your student ID to the test so the tutor can verify your identity.

This quiz was locked 19 Jun at 23:59.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	89 minutes	92 out of 100

❗ Correct answers are hidden.

Score for this quiz: 92 out of 100

Submitted 19 Jun at 10:03

This attempt took 89 minutes.



Question 1

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ Counters allow Mappers and Reducers to pass aggregate values back to the driver after the job has completed.
- ☒ The counter in MapReduce environment is composed of only a name and a value.
- ☐ The counter in MapReduce environment is composed of a class, a name, and the user-defined variables.
- ☐

Counters allow Mappers and Reducers to pass aggregate values back to the driver while the job is executing in real-time.

Week 05 - P39.



Question 2

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ MapReduce uses a single reducer in the program to gather all results together in the same place by default.
- ☒ With a single reducer, MapReduce collects all key-value pairs in sorted order.
- ☐ The number of reducers in MapReduce are predefined and cannot be customised.
- ☐ With a single reducer, MapReduce collects all key-value pairs in the same order as the data is read from the input.

Week 06 - P11.



Question 3

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ Apache Avro is portable across multiple languages.
- ☐ Apache Avro is limited to Java and Hadoop API only.
- ☐ Apache Avro is in complex file format.
- ☒ Apache Avro is a serialisation format which is a popular alternative to SequenceFiles.

Week 06 - Lecture - P39.

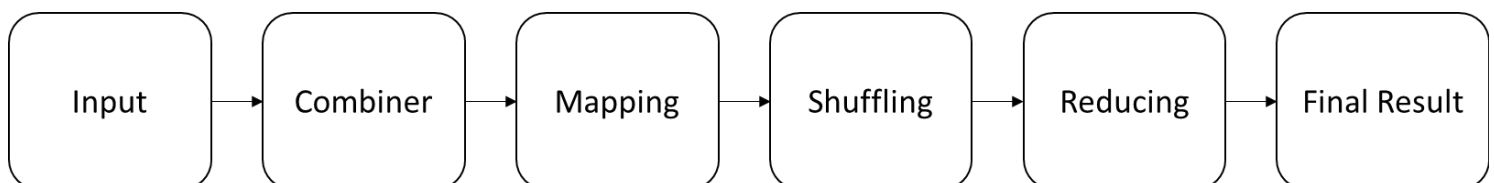


Question 4

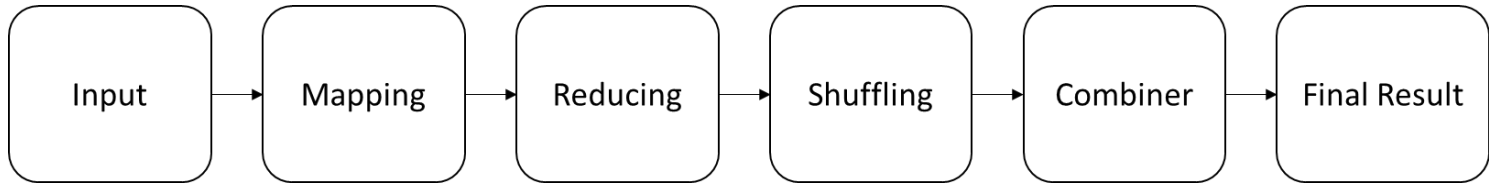
4 / 4 pts

Which of the following flowcharts regarding the MapReduce with combiner is correct?

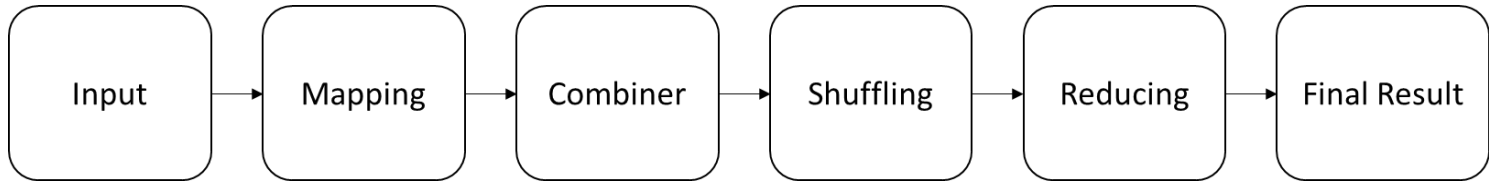
(a):



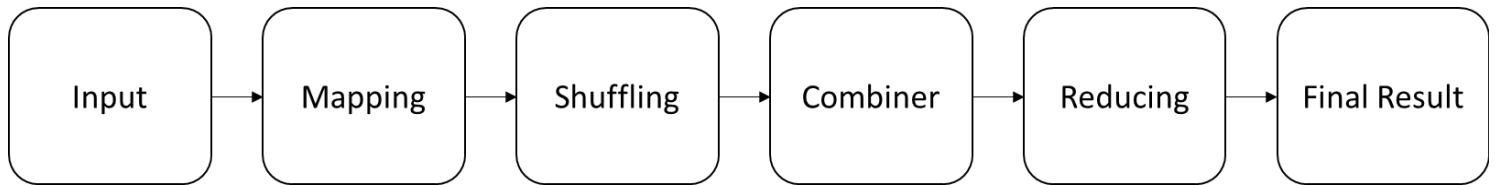
(b):



(c):



(d):



- ☐ (a)
- ☐ (b)
- ☒ (c)
- ☐ (d)

Week 04 - Lecture - P21.



Question 5

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ The Partitioner determines which Reducer each intermediate key-value pair goes to.
- ☐ The Partitioner determines which Combiner each intermediate key-value pair goes to.
- ☐ The default Partitioner in Hadoop is the LinearPartitioner.
- ☒ The default Partitioner in Hadoop is the HashPartitioner.

Week 06 - Lecture - P7, P9.

Incorrect



Question 6

0 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ ToolRunner can only be used by specifying configuration in the driver code.
- ☒ ToolRunner uses the GenericOptionsParser class internally.
- ☐ You can use ToolRunner in MapReduce driver classes.
- ☐ ToolRunner uses the GenericOptionsParser class externally.

Week 04 - Lecture - P5.



Question 7

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ Counters provide a way for Mappers or Reducers to pass aggregate values back to the driver after the job has completed.
- ☐ Counter has a variety of data types such as string and integers to be used in the code.
- ☒ Counter only has a name and a value as its variable types.
- ☐

Counters provide a way for Mappers or Reducers to pass aggregate values back to the driver while the job is executing.

Week 05 - Lecture - P41.



Question 8

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ It is recommended to test locally whenever possible when doing the debugging for MapReduce.
- ☐ It is recommended to avoid unit tests when debugging for MapReduce.
- ☒ It is recommended to make as much of your code as possible Hadoop-agnostic when debugging for MapReduce.
- ☐ It is recommended to start large and then narrow down the scope when debugging for MapReduce.

Week 05 - P5.

Incorrect



Question 9

0 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ The conf object has read in the Hadoop configuration files, and therefore knows the address of the NameNode.
- ☐ A file in HDFS is represented by a File object.

- ☒ A file in HDFS is represented by a Path object.
- ☐ You can use the .retrieve() method of the FileSystem object to get the instance of the configuration object.

Week 04 - Lecture - P27.



Question 10

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☐ You can use multiple reducers in the LocalJobRunner mode.
- ☒ You can only use a single reducer in the LocalJobRunner mode.
- ☐ The distributed cache works in LocalJobRunner mode.
- ☒ You can not use the distributed cache in LocalJobRunner mode.

Week 05 - Lecture - P9.



Question 11

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☐ The log4j library allows you to specify what gets logged, but when to log things can not be customised.
- ☒ The log4j library allows you to specify what gets logged and when to log things.
- ☐ The log4j library can only be called in the Mapper and the Reducer classes.
- ☒ The log4j library can be called in driver, mapper, and reducer classes.

Week 05 - Lecture - P23.



Question 12

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☒ The SequenceFiles are in binary formats and thus they have faster read/write speed than the text-formatted files.
- ☒ SequenceFiles are in binary formats.
- ☐ The SequenceFile is always readable regardless of the definition of the key or value object changes.
- ☐ The SequenceFiles are in binary formats and thus they have slower read/write speed than the text-formatted files.

Week 06 - Lecture - P38.



Question 13

4 / 4 pts

Match the items on the left to the descriptions on the right.

InverseMapper

Swaps keys and values. ▼

RegexMapper

Extracts text based on a regu ▼

LongSumReducer

Add up all values for a key. ▼

IntSumReducer

Add up all values for a key. ▼

Week 04 - Lecture - P35.



Question 14

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☐ When running the task using LocalJobRunner, it uses HDFS as the file system.
- ☒ When running the task using LocalJobRunner, it does not require any Hadoop daemons to be running.
- ☐ When running the task using LocalJobRunner, it requires the basic Hadoop daemons to be running.
- ☒ When running the task using LocalJobRunner, it uses the local file system instead of the HDFS.

Week 05 - Lecture -P7.



Question 15

4 / 4 pts

Which of the following description(s) is(are) correct?

- ☐ GZip is a splittable file format for compression.
- ☒ GZip is a non-splittable file format for compression.
- ☒ LZ4 is a splittable compression format under conditions.
- ☐ LZ4 is a splittable compression format with no limitations.

Week 06 - Lecture - P43.



The goal is to build a MapReduce-type program to analyse the log file.

We want to determine which IP address most frequently accesses our server in a specific month from the access log.

To answer this question, **please design a MapReduce program with proper mechanisms for gathering the desired information from the data and answer the following questions.**

The source log file name is "2022_access_log.gz", which is located on the path
"/home/training/training_materials/developer/data"

Hint 1: You can copy and paste an Eclipse project directly in the Package Explore, rename it into a new project, and then modify/create the corresponding contents in the project.

Hint 2: You may consider using a partitioner to help you achieve the goal. Importing the HashMap with the statements below may help.

```
import java.util.HashMap;
```

Hint 3: You may want to import the Arrays and List data types from Java to contain your data. The following statements can help you achieve that.

```
import java.util.Arrays;
```

```
import java.util.List;
```

Here is an example of creating a list from an array:

```
List<String> mylist = Arrays.asList("Apple", "Banana", "Onion");
```

Moreover, you may want to use the "sort" command with the "-k 2" and "-r" options in Linux to rearrange your output file for you. To put the sorted result in a new file, you can use the statement below:

```
sort -k 2 -r inputfile > outputfile
```

where -k 2 specifies the sort is based on the second column in the file, -r specifies the sort in descending order, and the ">" symbol stores the output on its left to the right as a file.

However, please note that the sort is applied sequentially from the first to the last digit. A larger number may be sorted to the later part in the list when a smaller number has a larger value in the first digit. For example, 68 will be sorted at the front than 256 because the first digit 6 is greater than 2.



Question 16

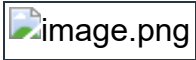
5 / 5 pts

What is the maximum connection count in August?

- ☐ 408
- ☐ 3708
- ☒ 8191
- ☐ 994

```
sort -k 2 -r part-r-00007 > Aug
```

```
less Aug
```



Question 17

5 / 5 pts

Which IP address(es) is(are) logged more than **400** connections in January?

Hint: To answer this question, you may use the grep command to help you.

Assume your input file contains:

a 300

b 200

c 100

d 50

The statement

grep -e 'c'

will give you the output as:

c 100

☐ 10.102.139.235

☐ 10.112.227.184

☒ 10.1.181.142

☒ 10.100.74.153

```
grep -e '10.1.181.142' part-r-00000
```

```
grep -e '10.100.74.153' part-r-00000
```



Question 18

5 / 5 pts

What is the maximum connection count in July?

☒ 20394

☐ 7289

☐ 8191

☐ 21588


```
sort -k2,2nr part-r-00006 | less
```



Question 19

5 / 5 pts

What is the connection count of IP 10.1.181.142 in February?

- ☒ 455
- ☐ 415
- ☐ 512
- ☐ 13675

```
training@quickstart:~/training_materials/developer/month_logs_match
File Edit View Search Terminal Help
10.1.181.142 455
10.1.182.140 3
10.1.186.241 10
10.1.189.131 1
10.1.189.181 3
10.1.189.198 1
10.1.190.77 15
10.1.192.99 1
10.1.193.151 1
10.1.194.115 1
10.1.194.73 1
10.1.195.150 2
10.1.20.205 1
10.1.200.62 2
10.1.202.148 1
10.1.203.155 1
10.1.203.32 1
10.1.204.33 2
10.1.21.186 2
10.1.210.1 1
10.1.212.93 24
10.1.214.142 1
10.1.214.174 1
:
```



Question 20

5 / 5 pts

Which IP address(es) is(are) logged more than **400** connections in March?

Hint: To answer this question, you may use the grep command to help you.

Assume your input file contains:

a 300

b 200

c 100

d 50

The statement

grep -e 'c'

will give you the output as:

c 100

☐ 10.89.223.120

☒ 10.6.34.78

☒ 10.247.240.153

☐ 10.174.2.172

grep -e '10.247.240.153' part-r-00002

grep -e '10.6.34.78' part-r-00002



Question 21

5 / 5 pts

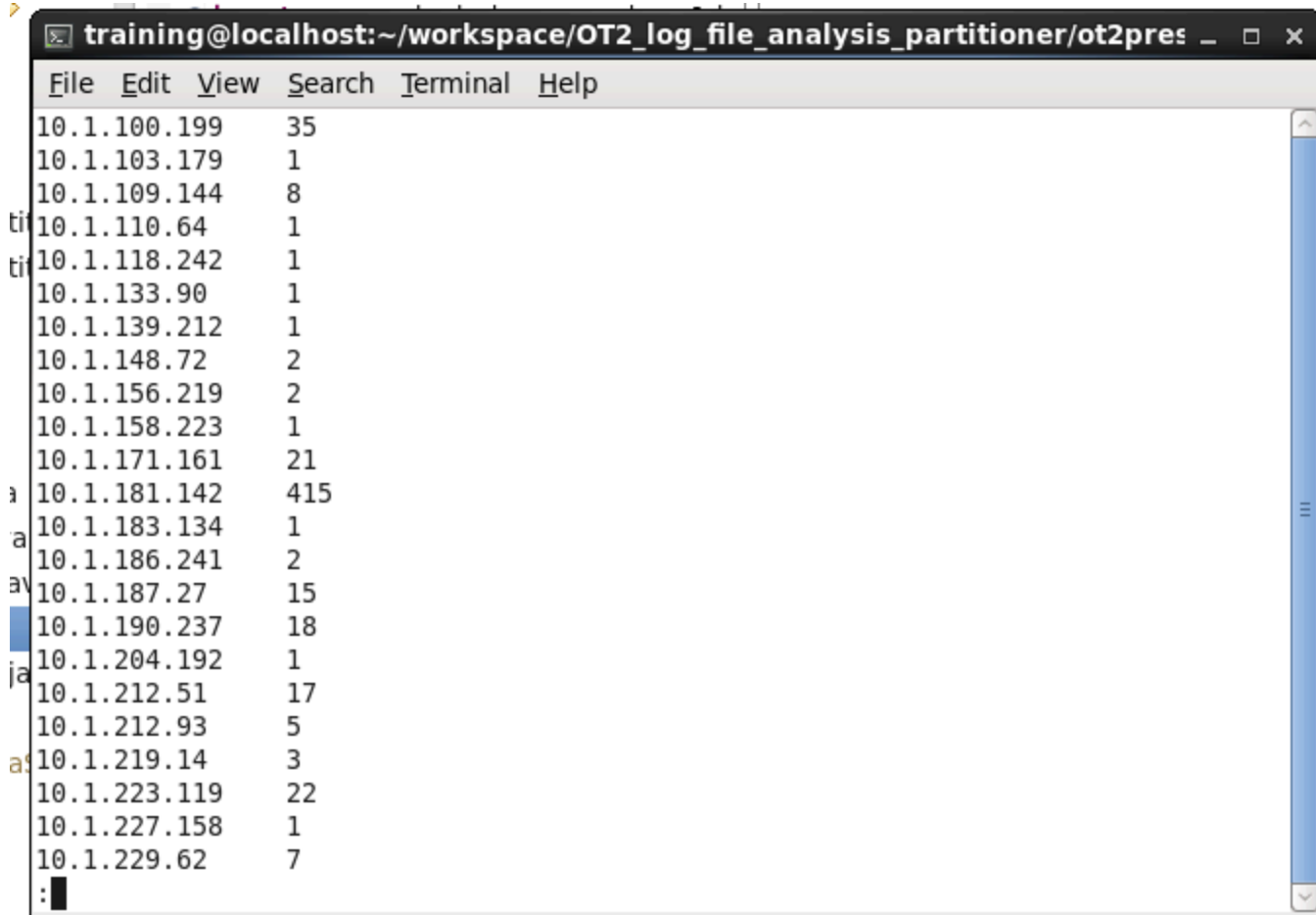
What is the connection count of IP 10.1.181.142 in January?

☐ 15

☐ 18

☐ 1423

☒ 415



IP Address	Count
10.1.100.199	35
10.1.103.179	1
10.1.109.144	8
10.1.110.64	1
10.1.118.242	1
10.1.133.90	1
10.1.139.212	1
10.1.148.72	2
10.1.156.219	2
10.1.158.223	1
10.1.171.161	21
10.1.181.142	415
10.1.183.134	1
10.1.186.241	2
10.1.187.27	15
10.1.190.237	18
10.1.204.192	1
10.1.212.51	17
10.1.212.93	5
10.1.219.14	3
10.1.223.119	22
10.1.227.158	1
10.1.229.62	7



Question 22

5 / 5 pts

Which IP address(es) is(are) logged more than **400** connections in February?

Hint: To answer this question, you may use the grep command to help you.

Assume your input file contains:

- a 300
- b 200
- c 100
- d 50

The statement

grep -e 'c'

will give you the output as:

c 100

☒ 10.108.132.137

☐ 10.138.58.93

☐ 10.73.68.117☒ 10.141.247.37

```
grep -e '10.108.132.137' part-r-00001
```

```
grep -e '10.141.247.37' part-r-00001
```



Question 23

5 / 5 pts

What is the maximum connection count in February?

☒ 13675☐ 8191☐ 15288☐ 4412

```
sort -k2,2nr part-r-00001 | less
```

Quiz score: 92 out of 100